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Fourth Semester B. Tech.

SUPPLIMENTARY EXAMINATION September 2019

HU202- Engineering Economics

Time: 3 Hours Max Marks: 50

Note: Answer any five questions.
Assume suitable Missing data, if any

1 a What do you mean by Price Elasticity of Demand? How it can be used in decision making. 2+3 =5

1 b i. If the demand law is given by $x=20/(p+1)$, find Elasticity of demand with respect to price at the point when $p = 3$. 2+3=5

ii. The demand x as a function of income y is given by $30x = 10+2y$. Obtain the expression for the income elasticity of demand and its value when $y = 250$.

2 a What do you mean by Recession? Suggest how engineers can bring the Indian economy out of recession. 5

2 b Investment proposals A and B have the net cash flows as follows: 5

	End of Year				
	0	1	2	3	4
A (Rs.)	-10,000	3,000	3,000	7,000	6,000
B (Rs.)	-10,000	6,000	6,000	3,000	3,000

The present worth of A with that of B at rate of interest = 18%. Which proposal should be selected?

3 a As a C.E.O. of a company, what measures you should take in deciding foreign location for your company? Discuss. 5

3 b The total cost function for producing a commodity in x quantity is $TC = 60-12x+2x^2$. 1+2+2=5

i. Find the average cost function

ii. The level of output at which this average cost function is minimum.

iii. Verify that AC=MC at the minimum point at AC curve.

4 a What do you mean by production process? Discuss changes which have taken place in the era of Information and Communication Technology? 5

4 b A company wants to set up a reserve which will help the company to have an annual equivalent amount of Rs. 10,00,000 for the next 20 years towards its employees' welfare measures. The reserve is assumed to grow at the rate of 15% annually. Find the single-payment that must be made now as the reserve amount. 5

5 a What do you mean by Monopolistic Competition? How it is different from Monopoly? 2+3=5

5 b A person is planning for his retried life. He has 10 more years of service. He would like to deposit Rs. 8.500 at the end of first year and thereafter, he wishes to deposit the amount with an annual decrease of Rs. 500 for the next 9 years with an interest rate of 15%. Find the total amount at the end of the 10th year of the above series. 5

6 a What do you mean by Central Bank? How it is different from Commercial Bank? 2+3= 5

6 b Two possible routes for laying a power line are under study, Data on the routes are as follows: 5

	Unit	Around the lake	Under the Lake
Length		15 Km	5 Km
First Cost	Rs.	1,50,000/Km.	7,50,000/Km.
Useful life	years	15	15
Maintenance cost	Rs.	6,000/Km./yrs	12,000/Km./yr
Salvage Value	Rs.	90,000/Km.	1,50,000/Km.
Yearly power loss	Rs.	15,000/Km.	15,000/Km.

If 15% interest is used, should the power line be routed around the lake or under the lake (use Annual Equivalent Method for analysis).

7 a Suggest five measures which can be taken for balancing environmental protection with economic development. 5

7 b A company must decide whether to buy machine A or Machine B. 5

	Machine A	Machine B
Initial Cost	Rs. 4,00,000	Rs, 8,00,000
Useful Life, in years	4	4
Salvage value at the end of machine life	Rs. 2,00,000	Rs. 5,50,000
Annual maintenance cost	Rs. 40,000	0

At 12% interest rate, which machine should be selected? (Use future Worth Method of Comparison).